

Background on causality

Background on causality

Consider the following graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$

$\mathcal{V}$: vertices, $\mathcal{E}$: edges

Path: $D \leftarrow B \rightarrow E \leftarrow C$

Directed path:

$A \rightarrow B \rightarrow E \rightarrow F \rightarrow G$

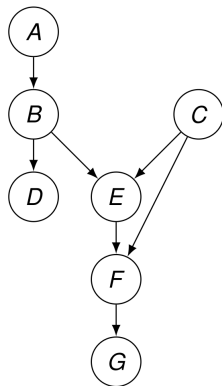
Parents: $Pa(E) = \{B, C\}$

Ancestors: $An(E) = \{A, B, C, E\}$

Children: $Ch(C) = \{E, F\}$

Descendants:

$De(C) = \{C, E, F, G\}$



Background on causality

Compatibility

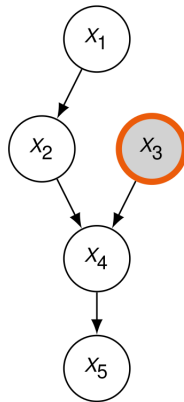
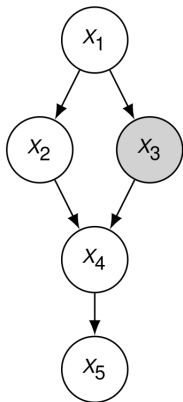
We say that a distribution $P(\mathcal{V})$ is compatible with (or Markov relative to) a DAG $\mathcal{G} = (\mathcal{V}, \mathcal{E})$

$$P(\mathcal{V}) = \prod_{X \in \mathcal{V}} P(X | Pa(X))$$

Factorization of $P(A, B, C, D, E, F, G)$ w.r.t. $\mathcal{G}$

$$\underbrace{P(A)}_{\text{no parents}} \cdot \underbrace{P(B | A)} \cdot \underbrace{P(C)}_{\text{no parents}} \cdot \underbrace{P(D | B)} \cdot \underbrace{P(E | B, C)} \cdot \underbrace{P(F | C, E)} \cdot \underbrace{P(G | F)}$$

Background on causality



Background on causality

Truncated factorization [Pearl 1995]

If we intervene on a subset $S \subset \mathcal{V}$, then

$$P(V_1 = v_1, \dots, V_d = v_d | do(S = s)) = \prod_{i \notin S} P(V_i = v_i | V_{pa_i})$$

if $v_1, \dots, v_d$ are values consistent with the intervention

Causal inference

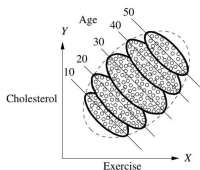
Definition

The causal effect $P(y|do(x))$ from a causal graph $\mathcal{G}$ is identifiable if $P(y|do(x))$ can be computed uniquely from observational data.

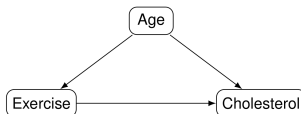
Aim of causal inference : identify effects

Causal inference

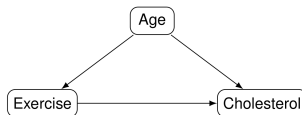
In a study, we measure weekly exercise and cholesterol levels for various age groups



What is the effect of exercise on cholesterol ?



Causal inference



$$\begin{aligned}P(a, e, c) &= P(a)P(e|a)P(c|a, e) \\P(a, c|do(e)) &= P(a)Pr(c|a, e) \\P(c|do(e)) &= \sum_a P(a)P(c|a, e)\end{aligned}$$

Causal inference

Theorem (identifiability in Markovian models)

Given a causal graph $\mathcal{G}$. We assume that the distribution $P(\mathcal{V})$ is compatible with this graph.

The causal effect $P(y|do(x))$ is identifiable whenever $\{X \cup Y \cup pa(X)\} \subset \mathcal{V}$, and is given by the direct causes adjustment:

$$P(y|do(x)) = \sum_{z \in pa(X)} P(y|x, z)P(z)$$

or if the parents set is empty

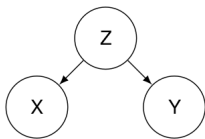
$$P(y|do(x)) = P(y|x)$$

Causal inference

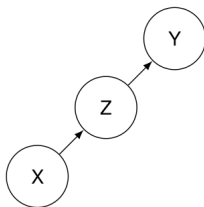
Beyond naive adjustment

- ▶ Is it possible to find a smaller adjustment set?
- ▶ Several criteria used to decide which choice of variables to adjust for to identify a causal effect.
- ▶ A popular one is backdoor based on the concept of d-separation

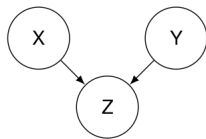
Causal inference



Fork



Chain



v-structure

The concept of d separation

Blocked paths

A path is said to be blocked by a set of vertices Z if:

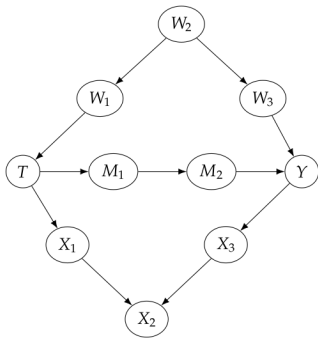
- ▶ it contains a chain $A \rightarrow B \rightarrow C$ or a fork $A \leftarrow B \rightarrow C$ and $B \in Z$, or
- ▶ it contains a collider $A \rightarrow B \leftarrow C$ such that no descendant of B is in Z

The concept of d separation

Definition

Two (sets of) nodes X and Y are d -separated by a set of nodes Z if all of the paths between (any node in) X and (any node in) Y are blocked by Z . We denote $X \perp\!\!\!\perp_G Y | Z$

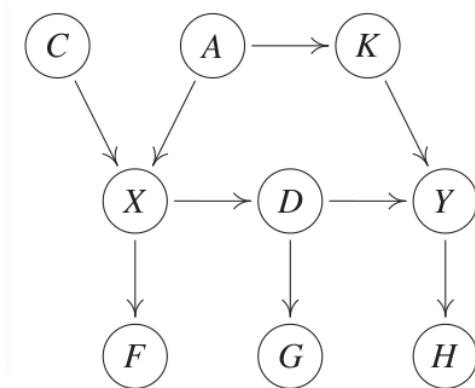
The concept of d separation



Are T and Y d separated by

- ▶ the empty set?
- ▶ $\{W_2\}$?
- ▶ $\{W_2, M_1\}$?
- ▶ $\{W_1, M_2\}$?
- ▶ $\{W_1, M_2, X_2\}$?

The concept of d separation



For this DAG : $C \perp\!\!\!\perp_G G | \{X\}$ and $C \not\perp\!\!\!\perp_G G | \{X, H\}$

Causal inference

The backdoor criterion

- ▶ A set Z satisfies the back-door criterion if:
 - ▶ Z contains no descendant of X , and
 - ▶ Z blocks every back-door path from X to Y , meaning every path that starts with an arrow into X ($X \leftarrow \dots$).
- ▶ If so, then

$$P(y|do(x)) = \sum_z P(y|x, z)P(z).$$

Causal inference

Beyond backdoor

- ▶ In whole generality use the three rules of Pearl's do calculus
- ▶ The ID algorithm

A case Study: Hotel Booking Cancellations

Example extracted from the DoWhy library

- ▶ Scientific question : estimate the impact of assigning a different room as compared to what the customer had reserved on Booking Cancellations.
- ▶ Gold standard : use experiments such as Randomized Controlled Trials wherein each customer is randomly assigned to one of the two categories i.e. each customer is either assigned a different room or the same room as he had booked before
- ▶ What if we cannot intervene or its too costly too perform such an experiment?
- ▶ Can we somehow answer our query using only observational data or data that has been collected in the past?

We shall answer these questions using causality and the dowhy library!

A case Study: Hotel Booking Cancellations

Description of the Dataset

- ▶ Booking information for a city hotel and a resort hotel taken from a real hotel in Portugal
- ▶ Includes information such as when the booking was made, length of stay, the number of adults, children, and/or babies, and the number of available parking spaces, among other things
- ▶ All personally identifying information has been removed from the data

Reference of the study case¹

¹<https://www.sciencedirect.com/science/article/pii/S2352340918315191>

Challenges and principles

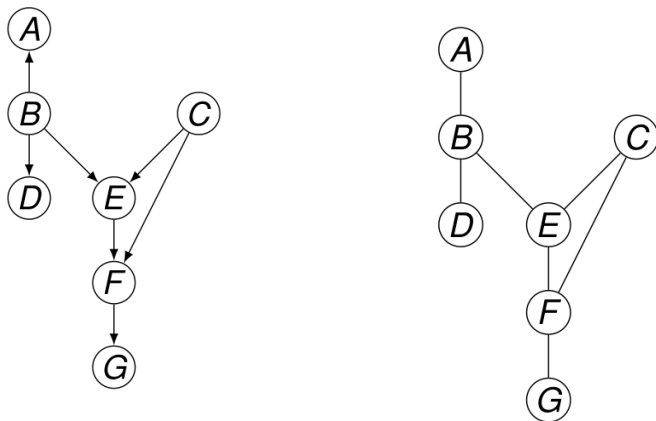
- ▶ In general, causal discovery from observational data is not possible.
- ▶ But it is possible under additional assumptions.
- ▶ Several approaches in the literature
 - ▶ **Constraint based methods** : run local tests of independence to create constraints on space of possible graphs.
 - ▶ **Noise based methods** : find footprints in the noise that imply causal asymmetry.
 - ▶ ...

Principles of constrain based methods

Main steps

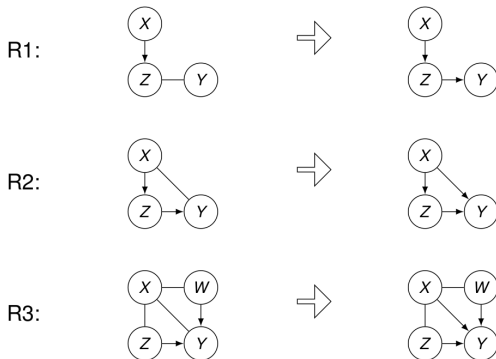
- ▶ Find **skeleton**
- ▶ Find **v-structures**
- ▶ Orient other edges using **basic rules**

Principles of constraint-based methods



a DAG and its corresponding **skeleton**

Principles of constraint-based methods



Basic rules

Principles of constraint-based methods

Algorithm 1 SGS

Input: $P(\mathcal{V})$

Output: CPDAG $\mathcal{G}^*$

- 1: Form the complete undirected graph $\mathcal{G}^*$ on vertex set $\mathcal{V}$
 - 2: **for** all $X - Y$ in $\mathcal{G}^*$
and subsets $S \subseteq \mathcal{V} \setminus \{X, Y\}$ **do**
 - 3: **if** $\exists S \subseteq \mathcal{V} \setminus \{X, Y\}$ such that $X \perp\!\!\!\perp_P Y \mid S$ **then**
 - 4: Delete edge $X - Y$ from $\mathcal{G}^*$
 - 5: **end if**
 - 6: **end for**
 - 7: **for** all $X - Z - Y$ in $\mathcal{G}^*$ such that $X \notin \text{Adj}(Y, \mathcal{G})$ **do**
 - 8: **if** $\nexists S \subseteq \mathcal{V} \setminus \{X, Y\}$ such that $Z \in S$ and $X \perp\!\!\!\perp_P Y \mid S$ **then**
 - 9: Orient $X \rightarrow Z \leftarrow Y$ in $\mathcal{G}^*$
 - 10: **end if**
 - 11: **end for**
 - 12: Recursively apply rules R1-R3 until no more edges can be oriented
 - 13: **Return** $\mathcal{G}^*$
-

The PC algorithm

- ▶ PC algorithm : optimized version of SGS
- ▶ Infer causal structure with the PC algorithm?
 - ▶ Infer mutual dependencies between variables : **skeleton of the causal graph**
 - ▶ Distinguish between causes and effects : **orientation of the v-structures of the causal graph**

Cause or consequence?

- ▶ Can we **distinguish** cause from effect?
- ▶ That is distinguish between these two causal graphs

$$X \rightarrow Y$$

or

$$Y \rightarrow X$$

using **observational data**.

Not always possible!

Cause or consequence?

The example of linear structural equation [f linear]

X cause Y if there exists $a \in \mathbb{R}$, ϵ^Y s.t.

$$Y = aX + \epsilon^Y, X \epsilon^Y.$$

Distinguish cause from consequence? [Shimizu et al., 2006]

Assume that $Y = aX + \epsilon^Y, X \epsilon^Y$ where all r.v. are continuous.
Then

$$\exists b \in \mathbb{R}, \epsilon^X \text{ s.t. } X = bY + \epsilon^X, Y \epsilon^X$$

iff (X, ϵ^X) are Gaussian random variables.

Existence of a non-linear extension of this result. 27/33

Noise based algorithm

Theorem (LiNGAM)

Assume a linear SCM with graph $G = (V, E)$ and a compatible distribution $P(V)$ such that for all $Y \in V$

$$Y = \sum_{X \in Pa(Y)} a_{xy} X + \xi_Y$$

where all ξ_Y are jointly independent and non-Gaussian distributed. Additionally, we require that for all $Y \in V$, $X \in Pa(Y)$, $a_{xy} \neq 0$. Then, the graph G is identifiable from $P(V)$.

Noise based algorithm

LINGAM

Algorithm 1 LINGAM

Input: $P(\mathcal{V})$

Output: $\mathcal{G}$

```

1: Form an empty graph  $\mathcal{G}$  on vertex set  $\mathcal{V} = \{X_1, \dots, X_p\}$ 
2: Let  $S = \{1, \dots, p\}$  and  $\mathcal{T} = []$ 
3: repeat
4:    $H = []$ 
5:   for  $i \in S$  do
6:     for  $j \in S \setminus \{i\}$  do
7:        $\hat{\xi}_{ij} = X_j - \frac{\text{cov}(X_i, X_j)}{\text{var}(X_i)} X_i$ 
8:     end for
9:      $h = \sum_{j \in S \setminus \{i\}} \lambda(X_i, \hat{\xi}_{ij})$ 
10:     $H = [H, h]$ 
11:   end for
12:    $i^* = \arg \min_{i \in S} H$ 
13:    $S = S \setminus \{i^*\}$ 
14:    $\mathcal{T} = [\mathcal{T}, i^*]$ 
15:    $\forall j \in S, X_j = \hat{\xi}_{i^*j}$ 
16: until  $|S| = 0$ 
17: Append( $\mathcal{T}, S_0$ )
18: Construct a strictly lower triangular matrix by following the order in  $\mathcal{T}$ , and estimate the connection strengths  $a_{i,j}$  by using some conventional covariance-based regression.
19: if  $a_{i,j} > 0$  then
20:   Add  $X_i \rightarrow X_j$  to  $\mathcal{G}$ 
21: end if
22: Return  $\mathcal{G}$ 

```

Noise based algorithm

Theorem (ANM)

Assume a linear SCM with graph $G = (V, E)$ and a compatible distribution $P(V)$ such that for all $Y \in V$

$$Y = f((X \in Pa(Y)) + \xi_Y$$

where all ξ_Y are jointly independent. Then, the graph G is identifiable from $P(V)$.

Noise based algorithm

ANM

Algorithm 2 ANM

Input: $P(\mathcal{V})$

Output: $\mathcal{G}$

```

1: Form an empty graph  $\mathcal{G}$  on vertex set  $\mathcal{V} = \{X_1, \dots, X_p\}$ 
2: Let  $S = \{1, \dots, p\}$  and  $\mathcal{T} = []$ 
3: repeat
4:    $H = []$ 
5:   for  $j \in S$  do
6:      $\hat{f}_j$ : Regress  $X_j$  on  $\{X_i\}_{i \in S \setminus \{j\}}$ 
7:      $\hat{\epsilon}_j = X_j - \hat{f}_j(X_i)$ 
8:      $h = \hat{\gamma}(\{X_i\}_{i \in S \setminus \{j\}}, \hat{\epsilon}_j)$ 
9:      $H = [H, h]$ 
10:  end for
11:   $i^* = \arg \min_{i \in S} H$ 
12:   $S = S \setminus \{i^*\}$ 
13:   $\mathcal{T} = [i^*, \mathcal{T}]$ 
14: until  $|S| = 0$ 
15: for  $j \in \{2, \dots, p\}$  do
16:   for  $i \in \{\mathcal{T}_1, \dots, \mathcal{T}_{j-1}\}$  do
17:      $\hat{f}_i$ : Regress  $X_i$  on  $\{X_k\}_{k \in (\mathcal{T}_1, \dots, \mathcal{T}_{j-1}) \setminus \{i\}}$ 
18:      $\hat{\epsilon}_i = X_i - \hat{f}_i(X_k)$ 
19:     if  $\{X_k\}_{k \in (\mathcal{T}_1, \dots, \mathcal{T}_{j-1}) \setminus \{i\}} \not\perp_P \hat{\epsilon}_i$  then
20:       Add  $X_i \rightarrow X_j$  to  $\mathcal{G}$ 
21:     end if
22:   end for
23: end for
24: Return  $\mathcal{G}$ 

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Bibliography

1. Pearl (1995). Causal diagrams for empirical research. Biometrika
2. Pearl (2009). Causality: Models, Reasoning, and Inference (2nd ed.). Cambridge University Press. See Ch. 3 for the backdoor and front-door criteria.

Bibliography

1. Spirtes et al. (1991). An algorithm for fast recovery of sparse causal graphs. Social Science Computer Review
2. Hoyer et al. (2009). Nonlinear causal discovery with additive noise models. NeurIPS